



BrightLearn
AI & DATA SKILLS

2026 EDITION · BEGINNER'S GUIDE

Data Stories

Turn raw data into clear visuals and compelling stories.

A complete beginner-to-advanced guide to data visualization, chart choice, insight writing, and data storytelling.

TOPIC

Visualization & Storytelling

SECTIONS

14 chapters

LEVEL

Beginner-friendly

INCLUDES

Charts · Insights · Checklists

Visualize · Interpret · Recommend



00

What is Data Visualization?

A spreadsheet full of numbers is not insight. Insight begins when you turn those numbers into a picture — and a picture into a story that helps someone make a decision.

DEFINITION

Data Visualization

The use of **visual elements — charts, graphs, maps, dashboards** — to communicate data clearly. Instead of asking people to read thousands of rows, you give them a picture that answers their question in seconds: what changed, what is winning, what needs attention.

★ WHAT GOOD VISUALIZATION HELPS PEOPLE DO

- **Understand data faster** than reading rows in a spreadsheet.
- **See patterns and trends** that hide in raw numbers.
- **Compare performance** across products, regions, channels, or time.
- **Detect outliers** — unusual values worth investigating.
- **Communicate findings** to non-technical audiences clearly.
- **Turn analysis into action** — the whole point of doing the work.

★ THE GOAL IS NOT "WHAT HAPPENED?"

Beginners treat visualization as a way to show what happened. Strong analysts use it to answer three questions at once: what happened, why does it matter, and what should we do next? Every guide and principle in this book exists to help you get from the first question to the third — confidently.



01 Why We Code Before We Visualize

Raw data is almost never ready for a chart. It is messy, incomplete, duplicated, inconsistent, and rarely in the shape you need. Code is how professionals reliably get from that mess to clean, repeatable analysis.

★ WHAT CODE LETS YOU DO RELIABLY

- **Cleaning** — fix missing values, remove duplicates, standardise spelling, correct data types, trim whitespace, fix wrong date formats.
- **Transforming** — extract month/year/quarter from dates, build age groups, calculate revenue, convert currencies.
- **Aggregating** — totals per month, averages per customer, counts per region — the summaries that make charts possible.
- **Repeating** — run the same logic next month, next quarter, next year without redoing the work by hand.
- **Reducing errors** — humans make mistakes; code applies the same rule every time.
- **Handling scale** — Excel struggles past a million rows. SQL, Python, Databricks, BigQuery handle billions.

A REAL CLEANING EXAMPLE

All four of these refer to the same province — but as written, a chart would show four "different" provinces:

101	·	Gauteng	·	500
102	·	gauteng	·	700
103	·	GP	·	300
104	·	Gauteng	·	900

Before any "Sales by Province" chart, code must standardise these to a single value: **Gauteng**.

★ THE TOOLS DATA PEOPLE USE

SQL for querying, filtering, joining, and aggregating data. Python for cleaning, automation, machine learning, and complex transforms. R for statistics. Excel for



quick checks. Power BI, Tableau, and Looker Studio for the dashboards. Databricks and BigQuery for the heavy lifting on big data. Each tool has its place; great analysts learn at least two and pick the right one for the job.



02

Exploratory Data Analysis

Before you build the final dashboard, you have to understand what is in the data. EDA is the deliberate, structured exploration that tells you what the data really looks like — and saves you from publishing a chart that turns out to be wrong.

DEFINITION

EDA

Exploratory Data Analysis is the process of **exploring a dataset before drawing conclusions** — checking its shape, distributions, quality, and relationships. It surfaces problems early, generates the first observations, and tells you which charts will actually be useful.

THE TWELVE CHECKS OF GOOD EDA

★ STRUCTURE & QUALITY

- **Dataset structure** — rows, columns, names, data types, example records.
- **Missing values** — what's missing, why, and what to do about it.
- **Duplicates** — duplicates inflate totals; check before any SUM or COUNT.
- **Data types** — dates as dates, numbers as numbers, IDs as strings if they should be.
- **Data quality** — negative sales, future dates, age 150, quantity 0, duplicate IDs.

★ PATTERNS & RELATIONSHIPS

- **Descriptive statistics** — mean, median, min, max, standard deviation, quartiles for every numeric column.
- **Distribution** — how values are spread (histograms, box plots).
- **Category frequency** — top and bottom categories, value counts (bar charts, treemaps).
- **Time trends** — sales by month, visits by hour, orders by weekday (line charts).
- **Relationships** — do two variables move together? (scatter plots, correlation heatmaps).



- **Outliers** — extreme values; real events, errors, or fraud signals?
- **Initial insights** — the first observations that will guide your final visuals.

★ WHY EDA MUST COME FIRST

Skip EDA and you risk publishing a chart that says "Gauteng vs gauteng vs GP" as three different regions, or a revenue total that double-counts duplicates, or a trend line that's distorted by a single outlier. EDA is the work that no one sees but everyone benefits from. Strong analysts treat it as non-negotiable.



03

EDA vs Presentation Visuals

Not every chart you make should be shown to the audience. The chart you create to understand the data is almost never the chart you create to communicate the message. Knowing which is which is a hallmark of a mature analyst.

EDA VISUALS

- Made for you — the analyst
- Can be messy, experimental, dense
- Purpose: find patterns, test ideas
- Show distributions and outliers
- Audience: yourself, your team
- Quantity over polish — make many

PRESENTATION VISUALS

- Made for the audience
- Must be clean, focused, polished
- Purpose: communicate one message
- Show the answer, not the search
- Audience: executives, clients, ops
- Quality over quantity — pick fewer

WORKED EXAMPLE — ORDER VALUE

EDA question:

"What does the distribution of order values look like?"

EDA visuals:

- Histogram of all order values across the dataset
- Box plot showing the spread and outliers

Presentation message:

"Most orders are below R500, but a small group of high-value customers drives most of the revenue."

Presentation visuals:

- Bar chart of customer value segments
- KPI card showing % of revenue from high-value customers



04 Charts — Comparisons & Trends

The most common questions in business analysis are "which is biggest?" and "how is it changing?" These six charts answer those questions — pick whichever fits the shape of your data.

BAR CHART

Best for: Comparing values across categories — sales by product, customers by province, revenue by store.

Use when: Category names are long, or you want easy comparison between groups. Sort highest to lowest.

COLUMN CHART

Best for: Same as bar, but vertical. Good for months, quarters, short labels.

Use when: Comparing values across short-labelled categories or short time periods (e.g. 12 months).

LINE CHART

Best for: Showing trends over time — monthly revenue, weekly visits, daily orders.

Use when: The x-axis is time and you want to show change, direction, or trend.

AREA CHART

Best for: Same as a line chart, but filled in. Emphasises magnitude and volume.

Use when: You want to show both the trend and the size of the values over time.

COMBINATION CHART

Best for: Two metrics on one chart — usually columns + a line on a second axis.

Use when: Two related metrics need to be compared together (e.g. revenue + profit margin).

STACKED BAR

Best for: A bar chart where each bar is split into segments by category.

Use when: Both the total and the composition matter (e.g. sales by region split by channel).



PITFALL — TOO MANY LINES

A line chart with twelve lines is a tangle, not a chart. **Keep it to three to five lines maximum.** If you need to compare more series, use small multiples (one chart per series) instead.



05

Charts — Composition & Distribution

These charts answer two different questions: "what is the breakdown of the whole?" and "how is this value spread out?" Mixing them up is one of the most common beginner mistakes.

PIE CHART

Best for: Parts of a whole — market share, channel mix, simple composition.

Use when: You have 2-5 categories and the differences are easy to see. Avoid otherwise.

DONUT CHART

Best for: Like a pie chart with a hole — good for dashboard summary cards.

Use when: 2-4 categories, with the key number placed in the centre.

TREEMAP

Best for: Hierarchical part-to-whole using rectangles — revenue by category and subcategory.

Use when: You want a single overview of contribution across a hierarchy.

100% STACKED BAR

Best for: Compares percentages across groups — totals always add to 100%.

Use when: Proportions matter more than totals (e.g. channel mix by region).

HISTOGRAM

Best for: Distribution of a numeric variable — order values, ages, delivery times.

Use when: You want to understand how a numeric value is spread out.

BOX PLOT

Best for: Distribution + median + quartiles + outliers across groups.

Use when: Comparing the spread of a numeric variable across categories.

PITFALL — PIE CHARTS WITH 15 SLICES

Pie charts collapse the moment you have more than five slices. **If you cannot tell the slices apart at a glance, switch to a sorted bar chart.** Never use 3D pie charts — they distort proportions for no benefit.



06

Charts — Relationships & Specials

The final group covers the questions "how do two things relate?", "where is this happening?", and a handful of charts that solve very specific business problems — funnels, waterfalls, Paretos, and the humble KPI card.

SCATTER PLOT

Best for: Relationship between two numeric variables — discount vs sales, age vs spend.

Use when: You want to explore whether two metrics move together (correlation, clusters).

BUBBLE CHART

Best for: Scatter plot + bubble size for a third variable.

Use when: Three measures matter at once (e.g. revenue, profit, and volume per product).

HEATMAP

Best for: Color intensity across a 2D matrix — sales by day & hour, correlations.

Use when: Showing patterns across two dimensions at once.

MAP

Best for: Geographic patterns — sales by province, customers by city.

Use when: Location is genuinely central to the story (not just available as a column).

KPI CARD

Best for: A single headline metric with context — total revenue, growth %, churn.

Use when: You need to lead with one important number on a dashboard or slide.

GAUGE CHART

Best for: Progress toward a target — sales achievement, SLA performance.

Use when: Sparingly — one important target at a time. They take up a lot of space.

FUNNEL CHART

Best for: Drop-off through a process — visits → product views → cart → purchase.

Use when: Analysing a step-by-step process where each stage filters down.

WATERFALL CHART

Best for: Positive and negative contributions to a final value — profit bridges.

Use when: Explaining how you got from one number to another (e.g. last month's profit to this month's).

PARETO CHART

TABLE



Best for: Bars + cumulative line — the 80/20 view.
Top items, top complaints.

Use when: You need to prioritize the small number
of items driving most of the result.

Best for: Exact details, row by row. Boring but
essential.

Use when: The audience needs the exact
numbers, not just the pattern.



07 Choosing the Right Chart

The fastest way to pick the right chart is to start with the question — not the data. Ask what you are trying to answer, and the chart practically picks itself.

THE DECISION TABLE

Question you are answering	Recommended chart
Compare values across categories	Bar chart, column chart
Show a trend over time	Line chart, area chart
Show parts of a whole	Pie, donut, treemap, stacked bar
Understand a distribution	Histogram, box plot
Explore a relationship	Scatter plot, bubble chart
Show patterns across 2 dimensions	Heatmap
Show geography	Map
Show progress to a target	KPI card, gauge
Show drop-off in a process	Funnel chart
Explain a change in a value	Waterfall chart
Show priority / 80-20	Pareto chart
Show exact values	Table

★ THE SEVEN-QUESTION CHART-CHOICE FRAMEWORK

- **What question am I answering?** The most important question — be specific.
- **Who is the audience?** Executive, manager, analyst, ops, marketing — each needs a different level of detail.
- **What decision will be made?** A chart with no decision behind it is decoration.
- **What kind of data is it?** Categorical, numerical, time-based, or geographic?
- **Is the question about comparison, trend, distribution, or relationship?** This narrows you to a small group of charts.
- **What is the main message?** The single takeaway the viewer should leave with.



- **What should they do after seeing it?** Action follows insight follows visual.



08 Data Storytelling

A chart on its own is just a picture. A data story is a chart wrapped in narrative — explaining what changed, why it matters, and what to do next. Storytelling is what turns analysis into business value.

DEFINITION

Data Story

The combination of **data, visuals, and narrative** that communicates a clear message. A good data story doesn't just show what happened — it explains why it matters and what should happen next.

THE THREE INGREDIENTS

Every data story has three ingredients — leave one out and it isn't a story.

01 • DATA

The facts, numbers, and evidence. "Revenue increased by 18% from March to April."

02 • VISUAL

The chart that makes the pattern easy to see. A line chart of revenue over six months.

03 • NARRATIVE

The explanation. "The increase was driven by higher demand in Electronics after the April promotion."

★ THE FIVE-PART STORY STRUCTURE

- **Context** — what are we looking at?
- **Observation** — what happened?
- **Explanation** — why did it happen?
- **Impact** — why does it matter?
- **Recommendation** — what should we do next?



09 Principles of Good Visualization

Eight principles separate visualizations that change minds from visualizations that get scrolled past. Internalise these and your charts will start landing with your audience every time.

01 CLARITY

A viewer should understand the chart in seconds. Clear title, readable labels, obvious message.

WEAK TITLE

- Sales by Category
- Monthly Revenue
- Q2 Numbers

STORY TITLE

- Electronics drove 42% of Q2 revenue
- Revenue grew 18% after April promo
- Gauteng outperformed all regions in Q2

02 SIMPLICITY

Remove anything that doesn't help the message. Avoid 3D effects, too many colours, heavy gridlines, unnecessary borders, and decorative icons that add nothing.

03 ACCURACY

Represent the data truthfully. Start axes at zero (or explain why not), don't hide important categories, and don't compare different time periods incorrectly.

04 CONTEXT

Numbers need context. "Revenue was R2 million" is weak. "Revenue was R2 million — 15% above target and 10% higher than last month" is strong.

05 FOCUS — ONE MESSAGE PER CHART



Every chart should have one main message. Ask: what should the viewer notice first? What is the takeaway? What decision does this support?

06 CONSISTENCY

Same colours for same categories. Same date and number formats. Same fonts. Consistency across a report builds trust; inconsistency erodes it.

07 ACCESSIBILITY

Readable font sizes, enough contrast, never rely on colour alone (use labels too), and avoid tiny text. About 1 in 12 men have some form of colour blindness — design for them.

08 AUDIENCE AWARENESS

★ DIFFERENT AUDIENCES, DIFFERENT NEEDS

- **Executive** — summary, impact, recommendation. Skip the methodology.
- **Manager** — trends, drivers, and what to act on this week.
- **Analyst** — details, methodology, data quality, edge cases.
- **Operations** — daily performance and exceptions.
- **Marketing** — customer segments and campaign results.



10

Presenting With Impact

A great chart can still die in a bad presentation. These eight rules — how to structure, lead, and deliver — are what get your analysis listened to in the room.

★ THE EIGHT RULES OF STRONG PRESENTATIONS

- **Start with the business question** — not the data. "How did sales perform, and what should we do?" keeps everyone focused.
- **Use a logical flow** — objective → data → findings → visuals → interpretation → risks → recommendations → next steps.
- **Lead with the answer** — executives want the headline first, the chart second. "Gauteng drove 42% of sales" beats "Here is sales by region."
- **One slide, one message** — never cram. One chart per slide if the chart is important.
- **Insight-based slide titles** — "Sales peaked in December due to festive demand" not "Monthly Sales".
- **Explain the why** — always offer a cause, not just the effect.
- **Connect insight to action** — every major finding needs a recommendation. Without action, insight is decoration.
- **Keep the audience with you** — clear visuals, short sentences, real examples, no jargon. Don't read every number on the slide.



11

Writing Strong Insights

An insight is not an observation. An observation says "sales went up." An insight explains what happened, by how much, why, why it matters, and what to do next. The difference is everything.

DEFINITION

The Insight Formula

Metric + Movement/Comparison + Driver + Business Meaning + Recommendation. Every part is required. If any one is missing, you've written an observation, not an insight.

THE FIVE QUESTIONS BEHIND EVERY INSIGHT

★ WALK THROUGH THESE BEFORE YOU WRITE

- **What happened?** — name the metric and movement clearly.
- **Compared to what?** — last month? target? last year? same period?
- **What caused it?** — propose at least one plausible driver.
- **Why does it matter?** — what's the business implication?
- **What should we do?** — every insight should end in a recommendation.



12 Bad vs Good Insights

The difference between an insight that gets ignored and one that drives a decision is rarely the analysis — it's the writing. Five before-and-after examples to anchor the formula.

EXAMPLE 01 — SALES TREND

X BAD INSIGHT

"Sales went up."

✓ GOOD INSIGHT

"Sales increased by 18% from March to April, mainly driven by a 25% lift in Electronics revenue. This suggests the April promotion was effective in high-value categories. The business should review the campaign audience and repeat similar promotions for Electronics customers."

EXAMPLE 02 — CUSTOMER SEGMENTS

X BAD INSIGHT

"Young customers buy more."

✓ GOOD INSIGHT

"Customers aged 25–34 generated 38% of total online revenue, making them the highest-value age group with a higher average order value than other groups. Marketing should prioritise this segment for personalised online campaigns."

EXAMPLE 03 — REGIONAL PERFORMANCE

X BAD INSIGHT

"Gauteng is the best province."

✓ GOOD INSIGHT

"Gauteng contributed 42% of total sales — more than double the second-highest province. Gauteng remains the strongest revenue market, but the business may be over-reliant on one region. Growth opportunities should be explored in Western Cape and KwaZulu-Natal."

EXAMPLE 04 — PRODUCT PERFORMANCE



X BAD INSIGHT

"Laptops sold well."

✓ GOOD INSIGHT

"Laptops were the top-performing product at R1.2 million in revenue — 22% of total Electronics sales. The strong performance is likely linked to back-to-school demand. The business should ensure stock availability before the next school season."

EXAMPLE 05 — FUNNEL DROP-OFF

X BAD INSIGHT

"Customers dropped off at checkout."

✓ GOOD INSIGHT

"The checkout stage had the highest drop-off — 62% of customers abandoned their cart before payment. This may point to friction in payment options, delivery fees, or the checkout flow. The business should review the checkout design and test payment incentives."



13

From Data to Decision

A complete analysis moves through four levels — data, observation, insight, recommendation. Here's what that looks like end-to-end on a real business question.

THE FOUR LEVELS OF ANALYSIS

LEVEL 1 • DATA

Total sales in April were R2.4 million.

LEVEL 2 • OBSERVATION

Sales increased by 18% compared to March.

LEVEL 3 • INSIGHT

The increase was mainly driven by Electronics, which grew 25% after the April promotion.

LEVEL 4 • RECOMMENDATION

Repeat the promotion for Electronics customers and test similar campaigns in Home Appliances.

A COMPLETE RETAIL STORY

Business question:

How did retail sales perform this quarter, and where should the business focus next?

The story:

Sales increased by 14% this quarter, mainly driven by Electronics and Home Appliances. Gauteng contributed the largest share of revenue, while KwaZulu-Natal showed the fastest growth. However, profit margin declined due to discount-heavy promotions. The business should continue promoting high-demand categories but review the discount strategy to protect profitability — and explore the growth opportunity in KwaZulu-Natal.

★ THE FINAL PROJECT CHECKLIST

- **Data prep** — loaded, columns understood, missing values checked, duplicates removed, types corrected, aggregations done.
- **EDA** — summary stats, distributions, categories, time trends, relationships,



outliers, quality issues documented.

- **Visualization** — right chart, clear title, labelled axes, formatted numbers, meaningful colour, uncluttered, supports the question.
- **Insight** — metric, comparison, business meaning, recommendation. Specific, not vague.
- **Presentation** — clear objective, logical flow, one message per slide, readable visuals, actionable recommendations, limitations and next steps stated.



From data to decision.

The best data analysts are great storytellers. Master these principles — clean data, the right chart, structured insights, and clear recommendations — and you'll get listened to.

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